A simulation of calculating and reducing CO_2-emissions from transport movements to customers in the Netherlands

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Abstract.

1. Introduction

Human activities such as the combustion of fossil fuels (coal, oil and gas) have caused the amount of the greenhouse gas, carbon dioxide (CO_2), to significantly increase in the atmosphere, especially over the last 50 years. More greenhouse gases retain more heat, which has caused global temperatures to increase in recent years. If this will continue, it will lead to a global change in the climate and might cause immense problems (*e.g.*, sea level rise and extreme weather events) (Houghton, 2012). To combat climate change, over 195 countries have signed the Paris Agreement in 2015. Its goal is to limit global warming to preferably 1.5 or even well below 2 degrees, as compared to pre-industrial levels. To achieve this long-term temperature goal, each country and their contributing sectors, must contribute (UNFCC, 2021).

The transport sector is the fourth largest emitter of greenhouse gases (after the industrial- and built environment sector). The sector is responsible for almost a quarter of Europe's greenhouse gas emissions and is the main cause of air pollution in cities (European Commission , 2021). Within this sector, road transport plays the most dominant role and projections indicate that by 2050, the demand for freight transport has tripled. Despite the foreseen growth in the sector, this does not necessarily have to mean an increase in CO_2 emissions. In order to achieve above mentioned climate goal and stay under current levels, the logistics sector must work together with governments and their customers, and take reducing measures (Greene & Lewis, 2018).

However, in order to discover the potential of the measures, companies must first identify their CO_2 emissions. This also applies to Royal Ahrend, a company that is specialized in delivering sustainable office lifestyles to customers worldwide. In 2018, a total turnover of 315 million euros was generated, the larger part of sales and turnover generated in the Netherlands (Office Republic, 2019). The majority of the production of their goods is supplied by Ahrend's production company in Sint-Oedenrode. The goods are transported from this production location to the customers. A large number of transport movements are required to transport these products. These movements are done by road, sea and air. Within these movements, road transport plays the most dominant role.

As mentioned previously, in order to reduce emissions, the company must first identify and quantify their CO_2-emissions. Therefore, an animation was made of the transport movements to customers in the Netherlands in the period of august 2019 till July 2020 in Jupyter. To add complexity, the rides to multiple, differently located customers were also taken into account. A total of 988 rides (totaling 4047 customers) were visualized and hereby the CO_2-emissions of these rides were calculated. A picture of the animation is given in figure 1. The blue dots represent the trucks that drove to the customers.

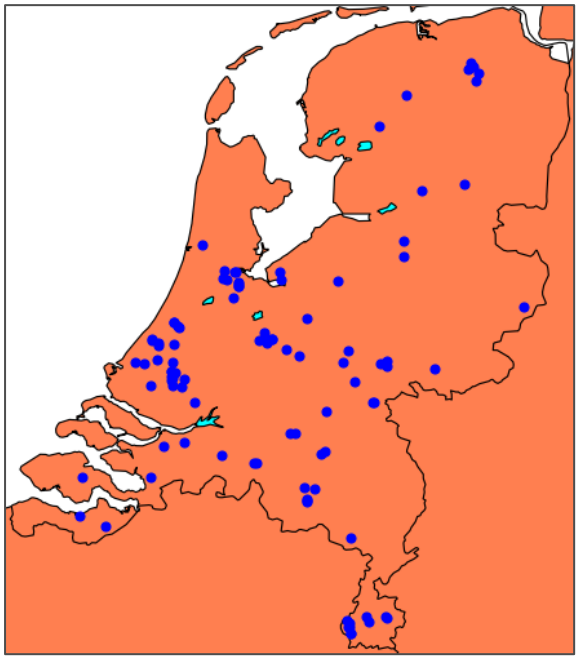

Figure 1 Animation rides Ahrend

The calculation of these CO_2-emissions was done according the NEN-EN 1628 standard called 'Methodology for calculation and declaration of energy consumption and GHG emissions of transport services (freight and passengers)'. This standard includes general principles, definitions, system boundaries, calculation methods, distribution rules and recommendations on which data can be useful for measuring greenhouse gas emissions (CEN, 2012). For this research, a specific focus was put on the principles that are important for road freight transport. Additionally, standard CO_2 emission factors for fuels are used for the calculation (CO2emissiefactoren, 2021).

After this, calculation, the potential of reducing the CO_2 emissions by changing the diesel trucks to electric trucks was explored. To add complexity, the place where they can charge was taken into account. Additionally, based on future developments regarding the range of electric trucks and the expansion of the charging infrastructure, multiple scenarios were simulated. These simulations looked at the influences of increasing the range and number of charging stations on the previously calculated CO_2 emissions. Also, the amount of trucks that could be replaced in the future by electric ones was evaluated.

2. Description of case study

The calculation of the CO_2 emissions is the first part and was used to calculate the emissions of each truck. Therefore, a list of the five most important parameters was made. This list is based on the characteristics from the NEN-EN 1628 standard. These parameters are:

1. The freight letter
2. Total distance (zip codes of each stop)
3. License plate of the truck
4. Fuel usage trucks
5. Order of the customers.

The freight letter (consisting of three digits and two letters) is used to sort the rides. Under each freight letter (e.g., 324PQ) the customer order, zip code of each customer, license plate of the truck that drove the freight letter (including the average fuel usage of that truck) can be found.

According to the NEN-EN standard, the various LEGs of a Vehicle Operation System (VOS) must be identified with associated activities. This is done according to the following steps:

- Identify the different LEGs of the transport service
- Calculate the CO2 emissions of each LEG
- Sum of the outcomes of each LEG (CEN, 2012)

For the transport of Royal Ahrend this would mean that the VOS can be divided into two legs, namely the main transport (LEG 1) whichrefers to the piece of transport from the production location in Sint-Oedenrode to the customers. The second LEG is called the after-transport andincludes the empty return kilometers. These are the kilometers from the last customer to the production location in Sint-Oedenrode. An example of the VOS (including the calculations) can be found in the figure 2.

Needed data:

- Total distance LEG 1: 98 km
- Total distance LEG 2: 20 km
- Average fuel usage truck: 4,46 km/l
- CO_2-emission factor diesel: 3,24 kg CO_2

Calculation CO2-emissions LEG 1 (3 customers)

1. Fuel usage: 98 km / 4,46 km/l = 22 l
2. CO2 LEG 1: 22 l * 3,24 kg CO_2 = 71,33 kg CO_2

Calculation CO2-emissions LEG 2 (back to factory)

1. Fuel usage: 20 km / 4,46 km/l = 4,48 l
2. CO_2 LEG 1: 4,48 l * 3,24 kg co2 = 14,5 kg CO_2

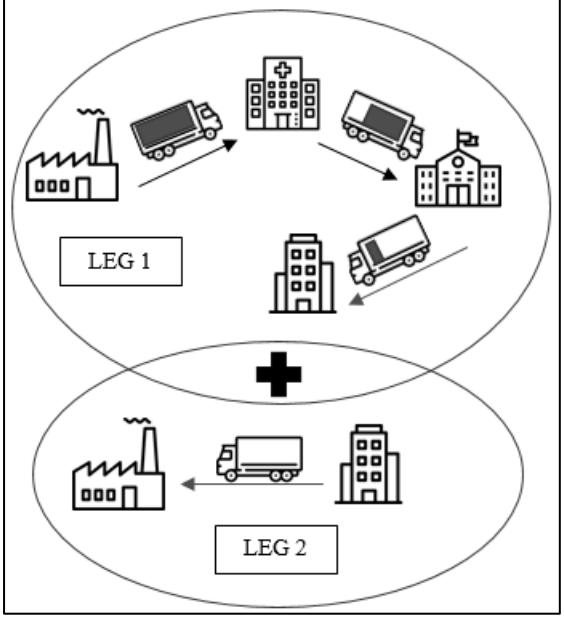

Figure 2. Example VOS and their calculation

The Great Circle Distance (GCD) is used to calculate the total distance of each ride. The GCD is in line with the NEN-EN standard and is the theoretically shortest distance between two points on the surface of the planet, measured along a path on the surface of the sphere (CEN, 2012). A postal code table with a representation of each zip code linked to the latitude and longitude was therefore created. The long- and latitude of the Netherlands postal codes was retrieved from AggData (AggData , 2021). A more detailed description of the application of the GCD can be found in chapter three.

The second part is a simulation that examines how these CO_2 emissions can be reduced due to the switch from diesel trucks to electric. At Royal Ahrend there is currently one electric truck, the DAF CF Electric with VDL E-power with a range of 100 km (Pals, 2019). To make the simulation as realistic as possible, this truck was used for the simulation. Also, the truck can currently only be charged on the dock of DHL in Eindhoven. In the simulation, multiple scenarios were developed by taking the following future developments into account:

A. Developments in the electric truck's radius

DAF trucks is taking the next step by making a new generation of batteries for the DAF CF Electric VDL E-power with a range of 200 kilometers (Velthoven, 2020). Therefore, the radius in the simulation was enlarged up to 200 km.

B. Developments in charging networks

According to Elaad, the biggest challenge is to set up the charging network among the European corridor network, as at the moment there is no charging network for trucks yet (Dijkhuizen, 2020). Figure 3 gives an overview of the superregional nodal points in the Netherlands corridor network.

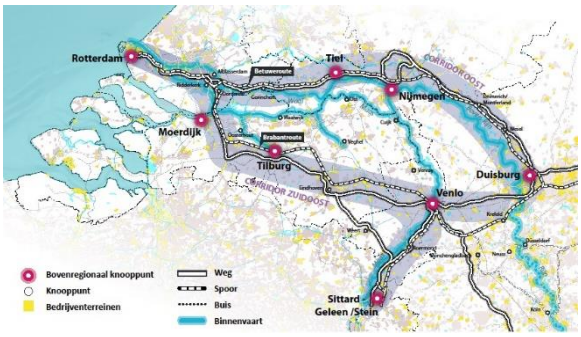

Figure 3. Corridor network of the Netherlands

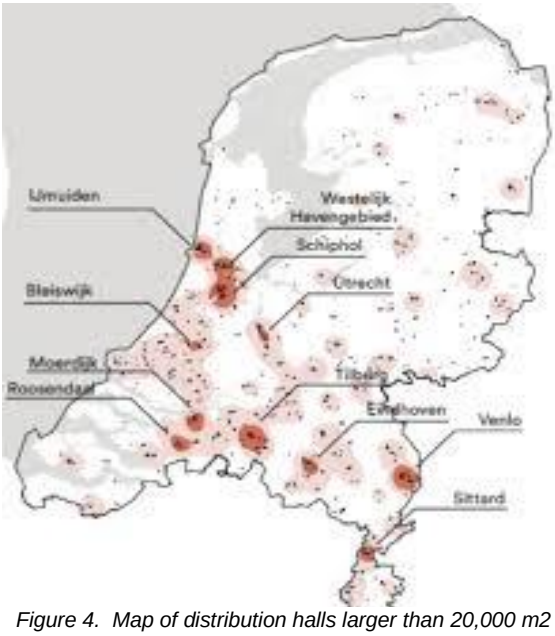

Figure 4. Map of distribution halls larger than 20,000 m2 (Natuur & Milieu , 2020)

The greatest demand for charging is expected to arise at large distribution halls on business parks (Natuur & Milieu , 2020). Figure 4 gives an overview of the largest distribution centers in the Netherlands.
Based on the map of the corridor network of the Netherlands, the map of the distribution halls and current developments is an estimate made of possible points for the charging network. The first possible place for a charging station in the simulation is in Geldrop (Eindhoven). According to multiple sources, this will be the first public charging station for e-trucks (Wijdeven, 2020).

Later in the simulation, three locations for extra charging stations were added. These locations were determined by comparing the map of the corridors and that of the large distribution halls (Figure 3 and 4, respectively). The points for the charging stations are: Western port area, Moerdijk and Sittard.

After taking future developments into account, a simulation was made to see the impact of these developments on the possibility to switch from diesel to electric (in %). More detail on the simulation will be explained in chapter 5.

3. Description of formulas

Description of the formulas that are involved in the system are discussed in this chapter. To calculate the distance between each point of the truck, the following formula is used (BlueMM, 2007):

$$=ACOS\Big(COS\big(RADIAND(90-Lat1)\big)*COS\big(RADIANS(90-Lat2)\big)+SIN\big(Radians(90-Lat1)\big)*SIN\big(Radians(90-Lat2)\big)*COS\big(RADIANS(Long1-Long2)\big)\Big)*6371$$

This formula calculates the distance between two latitude, longitude (lat/lon) points on the planet (BlueMM, 2007). As mentioned previously, for each zip code in the Netherlands, the matching latitude and longitude was found.
The latitude is known as the parallels and run in an east-west direction. The longitude is known as mederians and run in a north-south direction (Encyclopædia Britannica, Inc., 2021). An illustration of the latitude and longitude is given in figure 5. The 6371 (km) in the formula stands for the mean radius of the earth (BlueMM, 2007). A more detailed description of how the system should behave based on this formula is described in the following chapter.

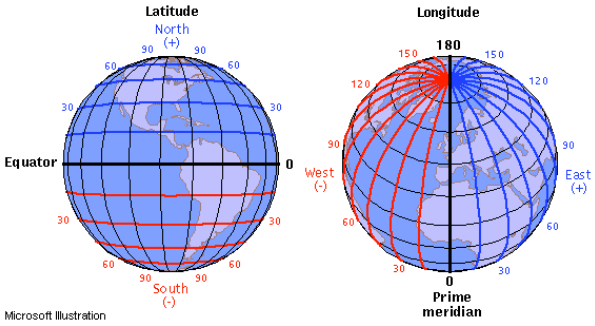

Figure 5 Latitude and longitude (Illinois State University, 2019)

After the distance between the stops of the truck was calculated, the following two formulas were added to calculate the CO_2 emissions (Engelen, 2014):

$$Use\ of\ the\ truck\ (l) = \frac{Number\ of\ kilometers\ (km)}{fuel\ usage\ of\ the\ truck\ (\frac{km}{l})}$$

$$Kg\ CO2e = Use\ of\ the\ truck\ (l) * emission\ factor\ CO2\ (kg\frac{CO2}{l})$$

4. Exploration of the system

Based on the three formulas that are discussed in the previous chapter the system should behave as follows:

- Based on the first formula, the system should calculate the shortest distance between each new position. So, if a truck goes from A to D, it should calculate the shortest possible distance from A to B, B to C and C to D.
- Based on the second formula, the system should use the total number of kilometers that were calculated by the first formula and divide it by the fuel usage of the truck that drove the freight letter.
- Based on the third formula, the system should calculate the CO_2 emissions in kilograms by multiplying the total amount of fuel usage (needed to drive the freight letter) with the CO_2 emission factor of the specific fuel that is used (3,24 for diesel) to deliver the orders to the customers.

Description of simulation

The purpose of this research was to provide insight in Ahrend's CO_2 emissions and to check the CO_2 reduction potential of implementing electric trucks. Therefore, data of previous transport movements to customers was used. Based on this data a simulation was made to

check which of these rides could have been replaced by an electric truck. For the transition to electric trucks, multiple scenarios were made based on enlarging the radius and the amount of charging stations.
A detailed level description of the simulation can be found in appendix A. The simulation starts with a check if the total distance of a specific ride was smaller than the radius of the electric truck. If this was the case, then the truck could immediately be replaced by an electric one.

If not, then a check was done which of the charging stations was the closest and check if it was possible to go to the station and the point after (by calculating the distance to the station and to the next point). Then, if the distance to the station plus the distance that was already driven was smaller than the radius (so that the truck is able to charge) **and** the distance to next stop plus the stops after were smaller than the truck's radius, the truck could also be replaced by an electric one. If this was not the case and there were still some orders left in the truck, the process started again until there were no point left. If there was, within all the points, no possibility to drive to the station and to the next points, the truck could not be replaced by an electric one.

5. Results of the simulation

First of all, the CO_2 emissions were calculated. The total amount of rides (988 rides) produced 185.000 kg CO_2. As mentioned previously, to make the scenarios as realistic as possible, future developments regarding the radius and charging stations were taken into account. The results of the simulations show how changing the radius and amount of stations influences the total CO_2 output. Also, the simulation provides the number of the total rides that could have been replaced by an electric truck (based on the different scenarios).

Scenario 1: One charging station

The first scenario (see Figure 6) shows the potential CO2 reduction with one charging station (=current situation). The charging station is the one at the DHL dock. Looking at the radius of the truck that is in possession of Ahrend (100 km) this would mean that a CO2 reduction of 6.900 kg (4%) was possible and that 15% of the 988 rides could have been replaced by the electric truck. The figure shows also that if the current battery of the electric truck will be replaced by the new one with a radius of 200 km, the CO2 reduction potential is 54.600 kg (29%) and around 50% of the rides could have been replaced by the electric truck.

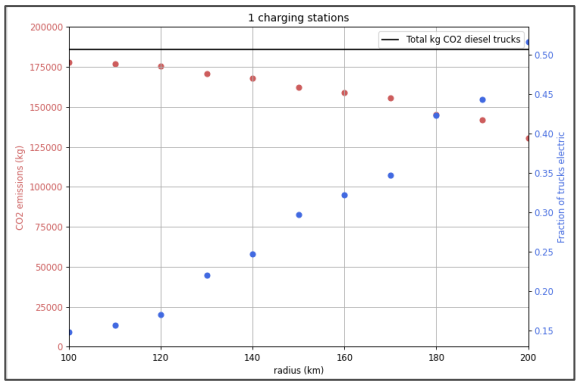

Figure 6 Scenario 1

Scenario 2: Two charging stations

The second scenario is the one with two charging stations. Now, there is the possibility to charge on the Dock of DHL and in Geldrop. As shown in Figure 7, it can be concluded that adding a charging station around 20 km from the first one does not have a significant influence on the CO2 emissions or the number of rides that could have been replaced by an electric one.

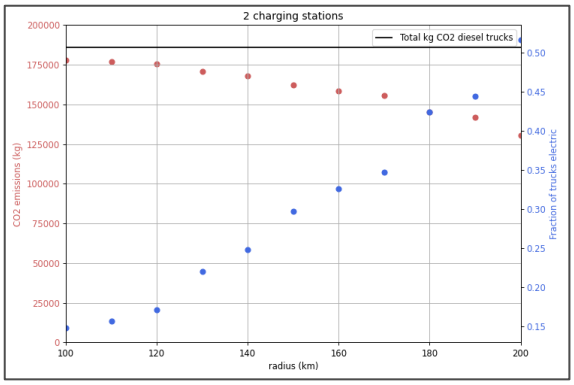

Figure 7 Scenario 2

Scenario 3: Three charging stations

In the third scenario, another charging station is added in the Western Docklands (near Amsterdam). Looking at the current radius of the electric truck, adding a station in Wester Docklands would not have made a difference in reducing Ahrend's CO_2 emissions. However, Figure 8 shows that it is interesting for Ahrend if the truck has a bigger radius. If the radius of the truck was 200 km, a CO2 reduction of 100.690 kg (54%) could have been achieved and 74% diesel trucks could have been replaced by

electric ones. Figure 8 shows the CO_2 reduction potential with three charging stations.

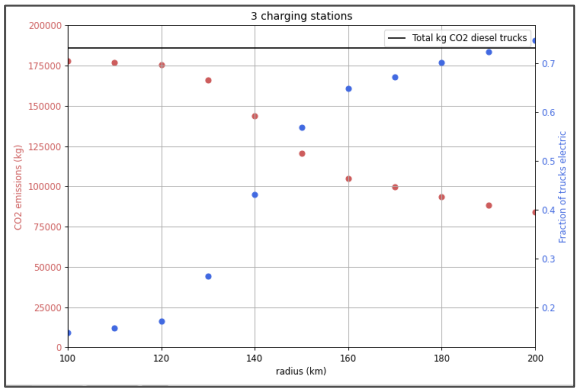

Figure 8 Scenario 3

Scenario 4: Four charging stations

In the fourth scenario, a fourth charging station is added along the Dutch corridor network on a distribution hall location in Moerdijk. This scenario shows that for the first time a CO2 emission reduction of more than 4% is possible when the electric truck has a radius of 100. A reduction of 12.340 kg CO_2 would have been possible (7%) and 20% of the rides could have been replaced. In addition, the graph shows little difference when the range enlarges as compared to the graph with 3 stations.

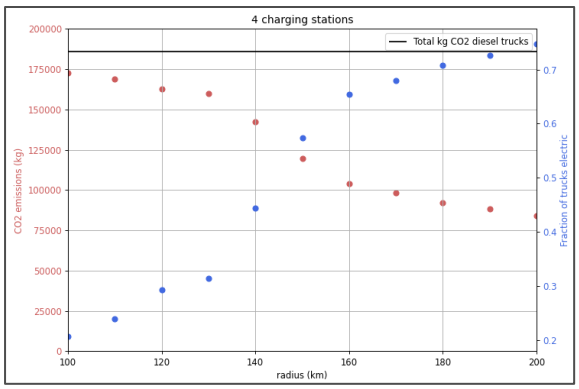

Figure 9 Scenario 4

Scenario 5: Five charging stations

In the last scenario, a fifth charging station was added in Sittard. For the transport of Ahrend this would have been interesting when the electric truck had a radius of 200 km. This would have led to replacement of 80% of the trucks to electric and a CO2 reduction of 61%.

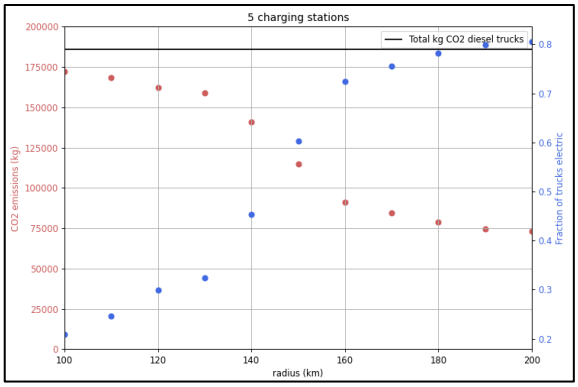

Figure 10 Scenario 5

6. Validation of the simulation

The following things were taken into account to make sure that the simulation approaches reality as much as possible.

6.1 Use of real data

Real time data was used to calculate the CO_2 emissions of the transport to customers. An example of this data can be found in appendix B. For the fuel usage of each truck, the average fuel sage of that specific truck was taken (based on 10 different rides). Secondly, the same customer order of the ride was used for the calculation of the distances, just like in real life.

6.2 Add a realism factor

Because the calculation of the CO_2 emissions are done by the use of the GCD, the shortest possible distance is taken between each point. In real life the truck drives across roads, therefore this cannot immediately be compared to the real distances driven. Therefore, a realism factor of 1.6 was added in the first formula. The realism factor of 1.6 is based on the comparison of 6 rides (3 big ones and 3 small ones) that are shown in Table 1. For the calculation of the real distance driven, Google Maps was used. The total distance of the freight letters that are mentioned in the table can be found in Appendix C.

Freight letter	Real distance driven (km)	GCD calculation python (km)	Difference
322PP (4 customers)	494	371	1,3
335PT (4 customers)	249	174	1,4
494PW (4 customers)	272	161	1,7
433PW (10 customers)	516	393	1,3
023PZ (9 customers)	331	193	1,7
475PR (12 customers)	308	135	2,2
Realism factor			1,6

Table 1 Determination realism factor

6.3 Research on future developments

In fact, the original idea of this simulation was to also include LNG trucks (as Ahrend has one LNG truck). But after calculating the radius of the LNG truck (1040 km), it was not relevant anymore for a simulation, as then every ride could have been replaced by the LNG one. Therefore, use of LNG trucks was excluded. Also, hydrogen trucks were excluded as the climate agreement of the Netherlands stated that there will only be 3000 trucks available around the year 2025 and the price of a hydrogen truck is 5 times as high as a diesel one (Klimaatberaad, 2018) (Wee, 2019).

So, this simulation looked explicitly at the potential of measurements that are interesting on short-term basis. To make sure that the outcomes of these simulations can be used by the company in the future, only the truck that currently is in possession of Ahrend was used. Secondly, realistic changes such as replacing the battery to enlarge the radius was also taken into account. Thirdly, location of charging places was based on literature research.

7. Simulation improvements

As mentioned previously, a standard value for each truck (based on 10 rides) is used for the determination of the fuel usage of the trucks. Therefore, the weight of the orders and the driving behavior of the driver were not taken into account. Secondly, although a realism factor was added in the simulation to get as close to reality as possible, a small difference might still be apparent.

Another limitation of this research is that not all possible CO_2 reduction measures have been calculated (*e.g.*, optimize transport planning, use of bio-diesel and modal shift). In future research, the CO2 reduction potential of other measurements should also be calculated. Regarding the measure of the electric truck, the amount of trucks that could have been replaced was based on position and not on available time. This could have influenced the real replacement of diesel by electrical trucks, because multiple rides were on the same day and Ahrend only possesses one electrical truck. As a result, the presented numbers in different scenarios might not be achievable at this moment.

Furthermore, in future simulations more complexity could be added by dividing the CO_2 emissions equally across the customers. The report of Panteia provides detailed information on how to achieve this (Engelen, 2014). Lastly, this research unfortunately does not address the CO_2 emissions caused by road transport to international customers. For future simulations it could be interesting to see which CO2 reduction measurements have the biggest potential regarding the international transport.

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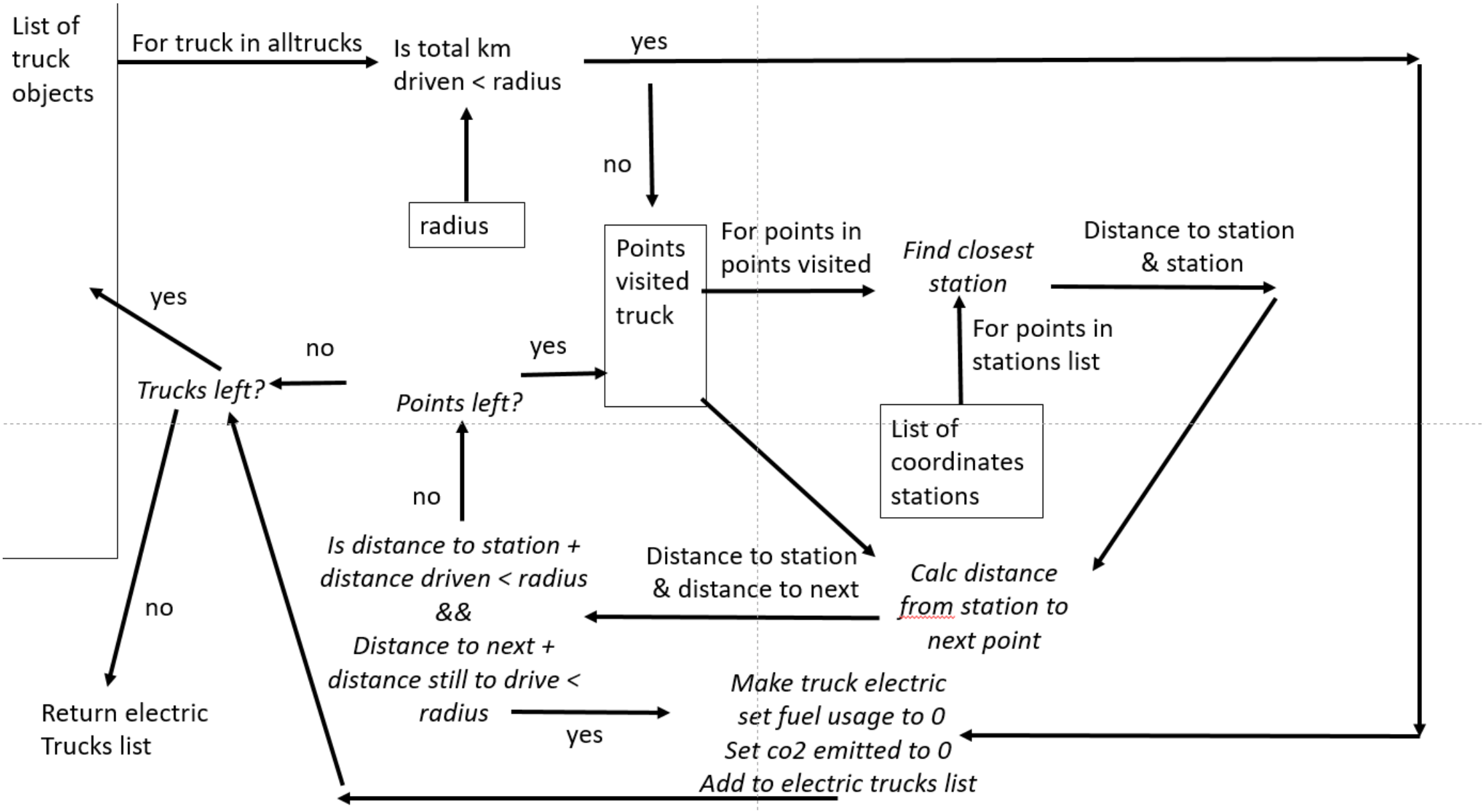
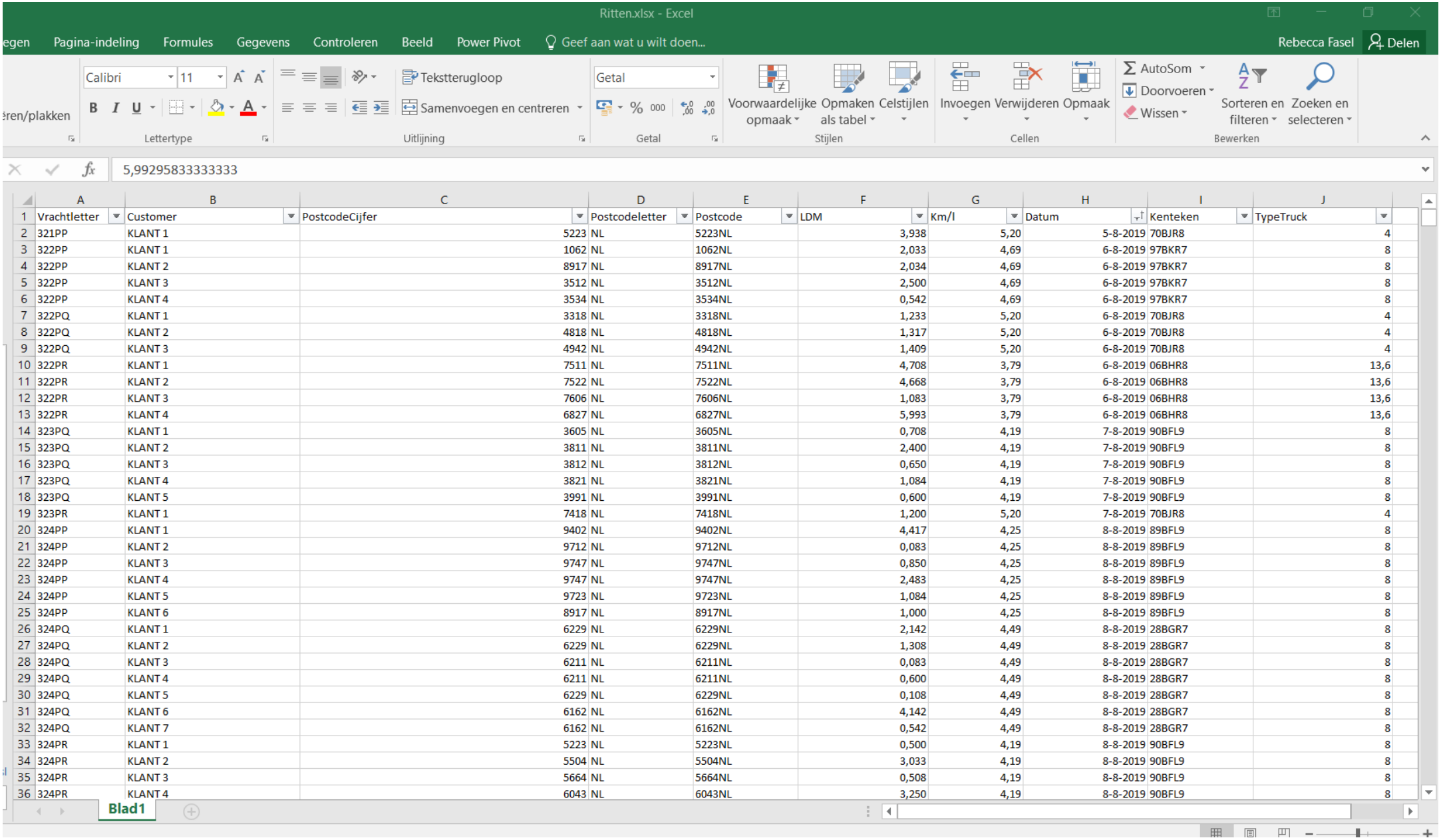
Appendix A. Detailed diagram of the simulation



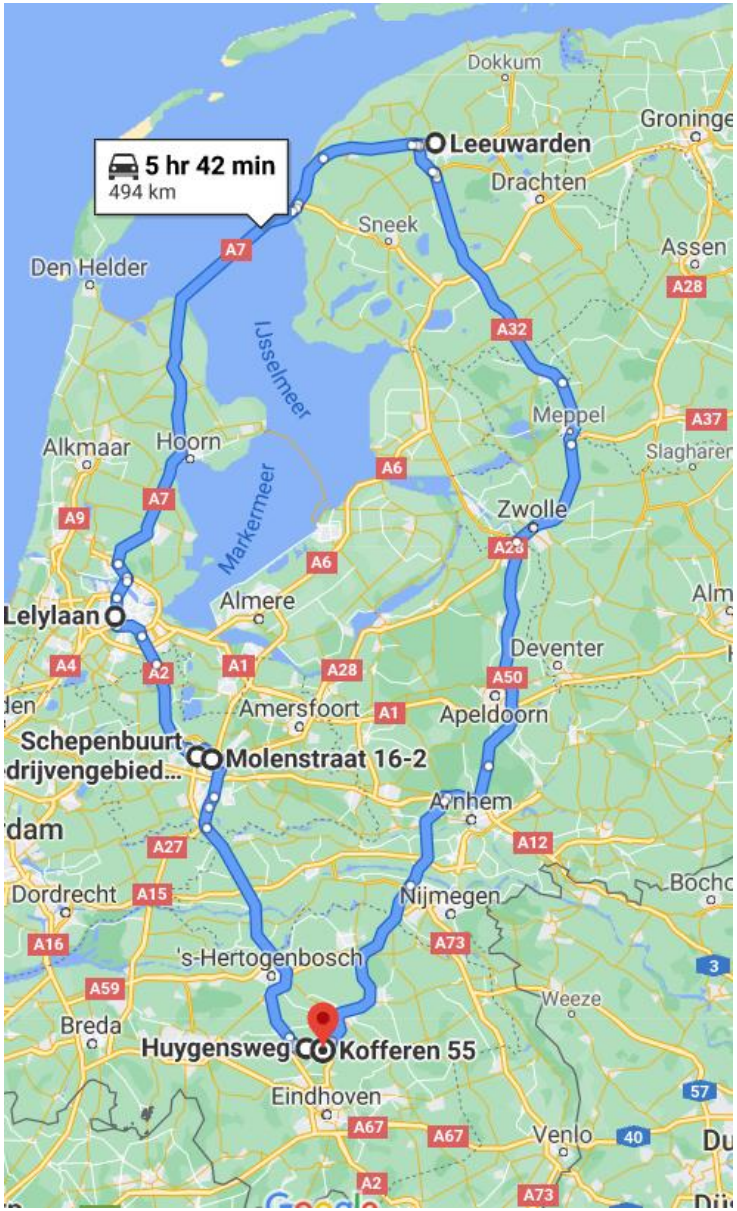
Appendix B. Example of the data

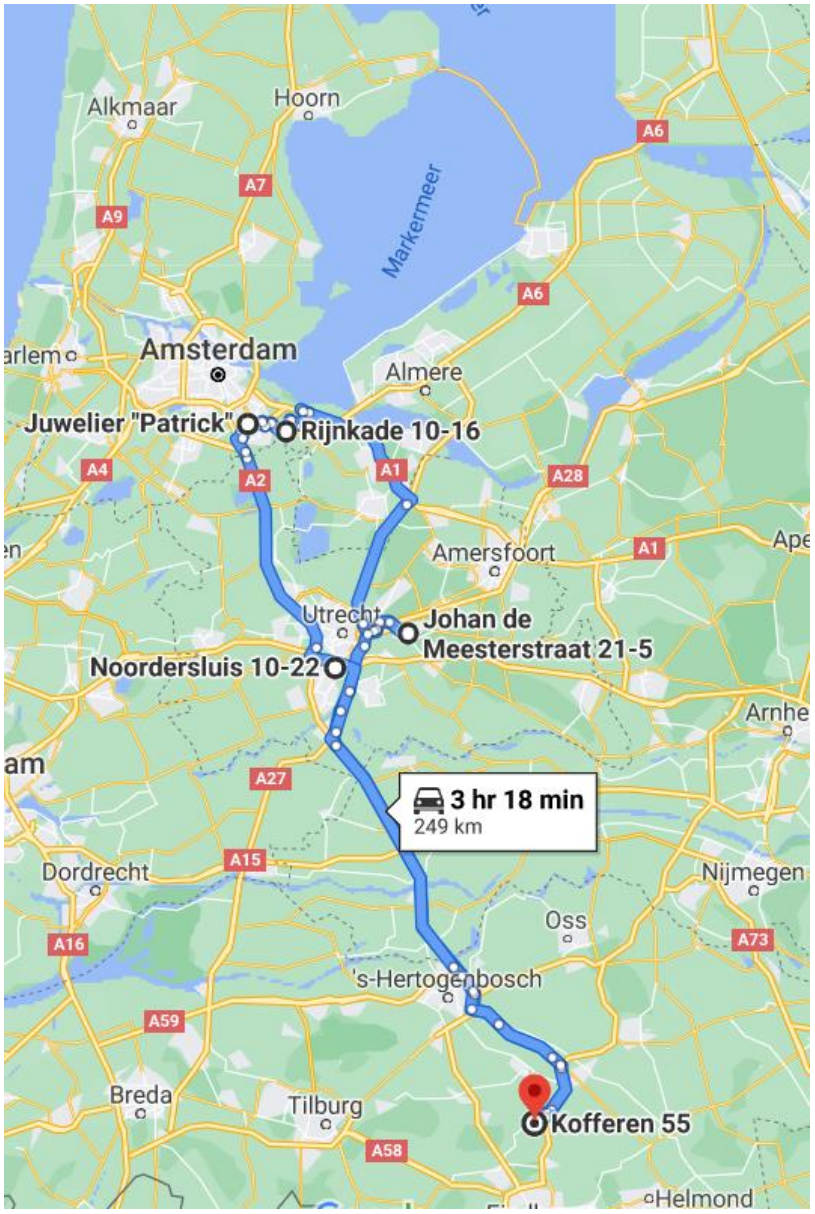


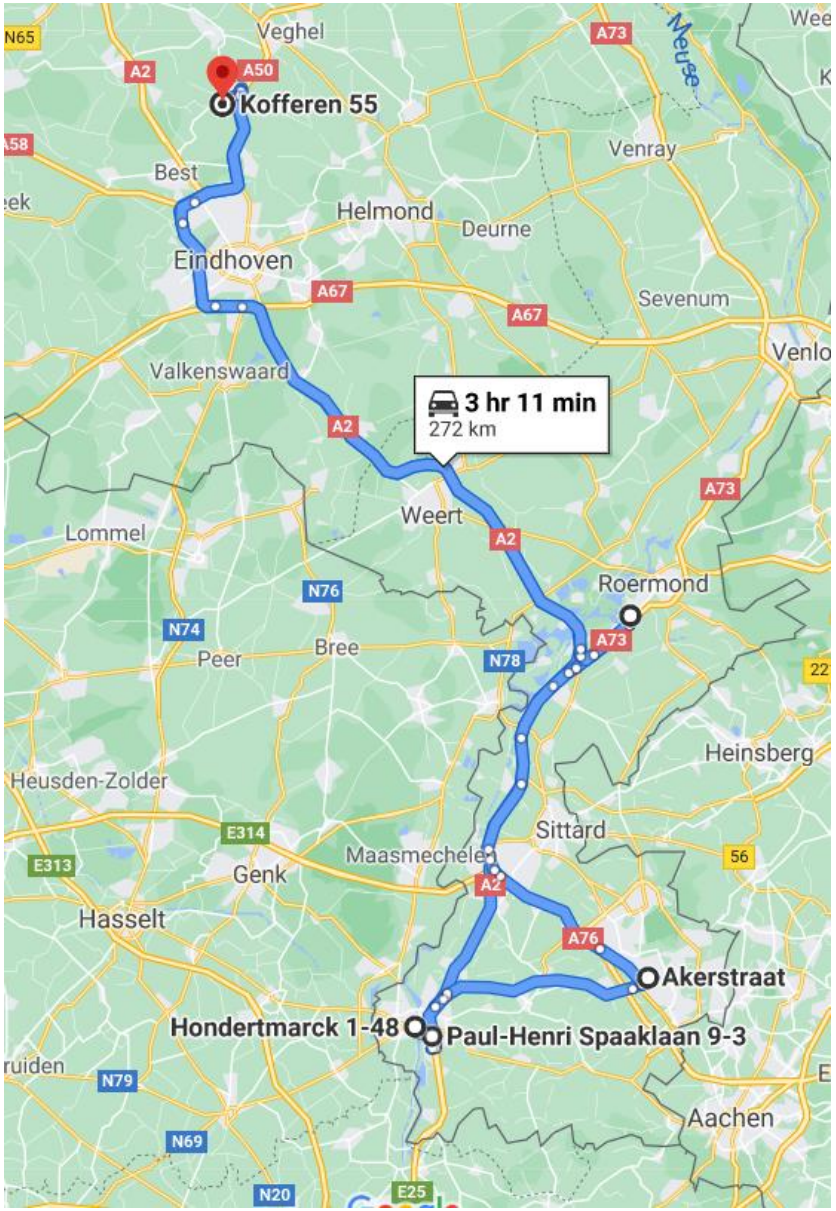
5,99295833333333

	A	B	C	D	E	F	G	H	I	J
1	Vrachtletter	Customer	PostcodeCijfer	Postcodeletter	Postcode	LDM	Km/l	Datum	Kenteken	TypeTruck
2	321PP	KLANT 1	5223	NL	5223NL	3,938	5,20	5-8-2019	70BJR8	4
3	322PP	KLANT 1	1062	NL	1062NL	2,033	4,69	6-8-2019	97BKR7	8
4	322PP	KLANT 2	8917	NL	8917NL	2,034	4,69	6-8-2019	97BKR7	8
5	322PP	KLANT 3	3512	NL	3512NL	2,500	4,69	6-8-2019	97BKR7	8
6	322PP	KLANT 4	3534	NL	3534NL	0,542	4,69	6-8-2019	97BKR7	8
7	322PQ	KLANT 1	3318	NL	3318NL	1,233	5,20	6-8-2019	70BJR8	4
8	322PQ	KLANT 2	4818	NL	4818NL	1,317	5,20	6-8-2019	70BJR8	4
9	322PQ	KLANT 3	4942	NL	4942NL	1,409	5,20	6-8-2019	70BJR8	4
10	322PR	KLANT 1	7511	NL	7511NL	4,708	3,79	6-8-2019	06BHR8	13,6
11	322PR	KLANT 2	7522	NL	7522NL	4,668	3,79	6-8-2019	06BHR8	13,6
12	322PR	KLANT 3	7606	NL	7606NL	1,083	3,79	6-8-2019	06BHR8	13,6
13	322PR	KLANT 4	6827	NL	6827NL	5,993	3,79	6-8-2019	06BHR8	13,6
14	323PQ	KLANT 1	3605	NL	3605NL	0,708	4,19	7-8-2019	90BFL9	8
15	323PQ	KLANT 2	3811	NL	3811NL	2,400	4,19	7-8-2019	90BFL9	8
16	323PQ	KLANT 3	3812	NL	3812NL	0,650	4,19	7-8-2019	90BFL9	8
17	323PQ	KLANT 4	3821	NL	3821NL	1,084	4,19	7-8-2019	90BFL9	8
18	323PQ	KLANT 5	3991	NL	3991NL	0,600	4,19	7-8-2019	90BFL9	8
19	323PR	KLANT 1	7418	NL	7418NL	1,200	5,20	7-8-2019	70BJR8	4
20	324PP	KLANT 1	9402	NL	9402NL	4,417	4,25	8-8-2019	89BFL9	8
21	324PP	KLANT 2	9712	NL	9712NL	0,083	4,25	8-8-2019	89BFL9	8
22	324PP	KLANT 3	9747	NL	9747NL	0,850	4,25	8-8-2019	89BFL9	8
23	324PP	KLANT 4	9747	NL	9747NL	2,483	4,25	8-8-2019	89BFL9	8
24	324PP	KLANT 5	9723	NL	9723NL	1,084	4,25	8-8-2019	89BFL9	8
25	324PP	KLANT 6	8917	NL	8917NL	1,000	4,25	8-8-2019	89BFL9	8
26	324PQ	KLANT 1	6229	NL	6229NL	2,142	4,49	8-8-2019	28BGR7	8
27	324PQ	KLANT 2	6229	NL	6229NL	1,308	4,49	8-8-2019	28BGR7	8
28	324PQ	KLANT 3	6211	NL	6211NL	0,083	4,49	8-8-2019	28BGR7	8
29	324PQ	KLANT 4	6211	NL	6211NL	0,600	4,49	8-8-2019	28BGR7	8
30	324PQ	KLANT 5	6229	NL	6229NL	0,108	4,49	8-8-2019	28BGR7	8
31	324PQ	KLANT 6	6162	NL	6162NL	4,142	4,49	8-8-2019	28BGR7	8
32	324PQ	KLANT 7	6162	NL	6162NL	0,542	4,49	8-8-2019	28BGR7	8
33	324PR	KLANT 1	5223	NL	5223NL	0,500	4,19	8-8-2019	90BFL9	8
34	324PR	KLANT 2	5504	NL	5504NL	3,033	4,19	8-8-2019	90BFL9	8
35	324PR	KLANT 3	5664	NL	5664NL	0,508	4,19	8-8-2019	90BFL9	8
36	324PR	KLANT 4	6043	NL	6043NL	3,250	4,19	8-8-2019	90BFL9	8

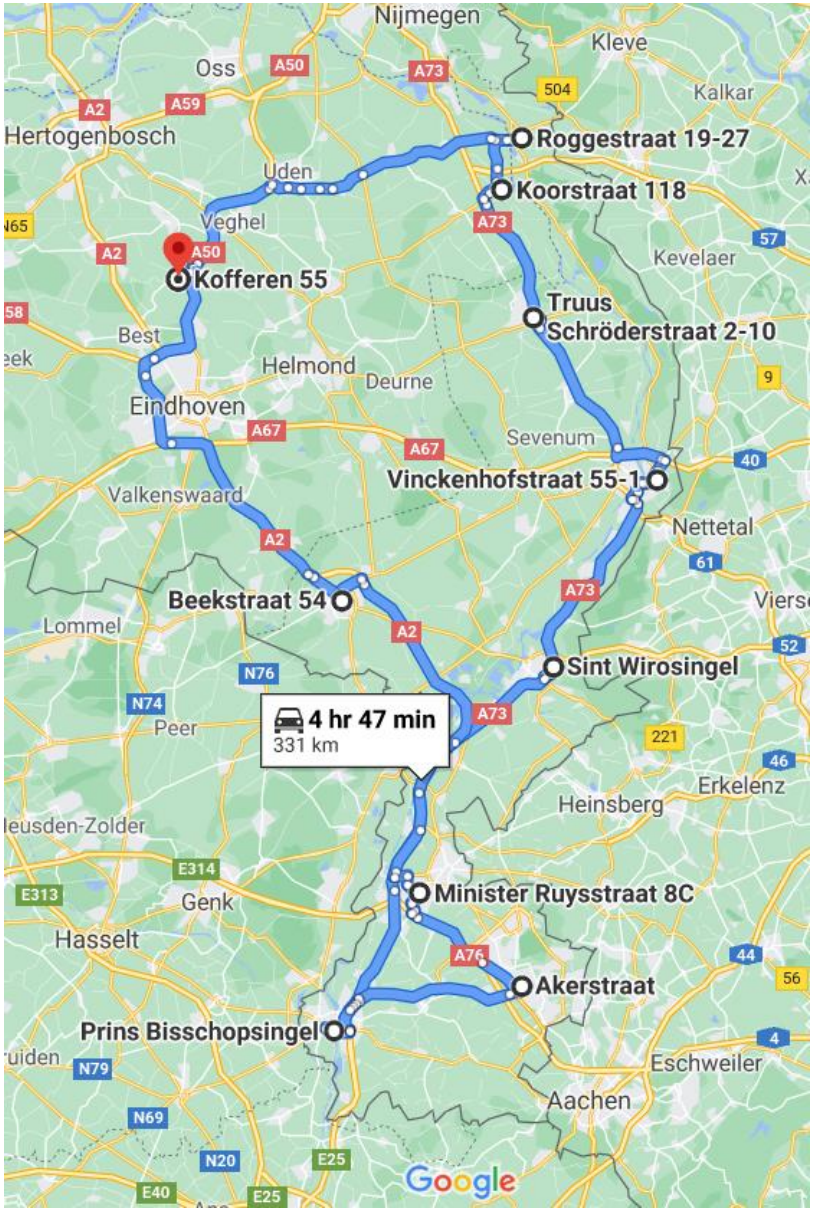
Appendix C. Real km driven

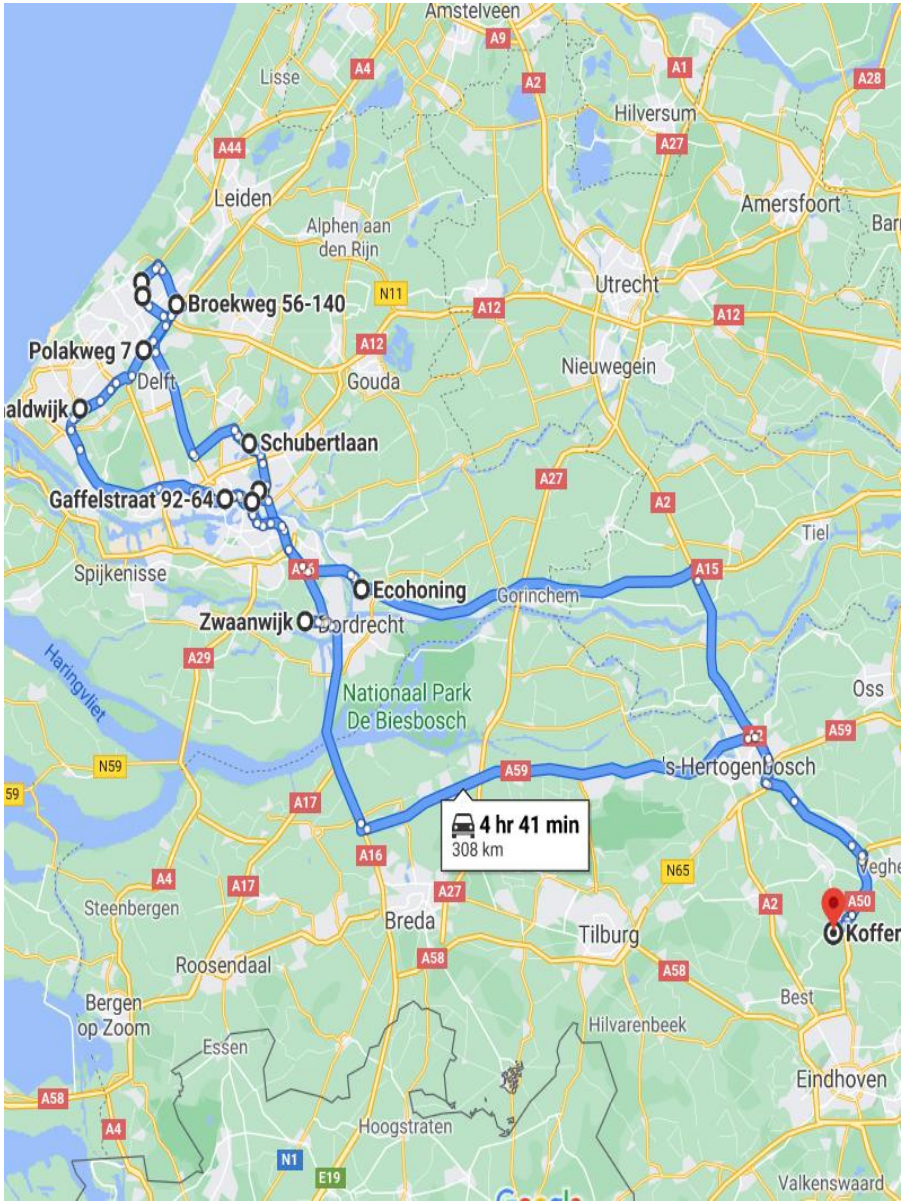





Kofferen 55
3 hr 11 min
272 km
Akerstraat
Hondertmarck 1-48
Paul-Henri Spaaklaan 9-3
Veghel
Best
Eindhoven
Helmond
Deurne
Venray
Sevenum
Valkenswaard
Weert
Lommel
Roermond
Peer
Bree
Heusden-Zolder
Heinsberg
Sittard
Maasmechelen
Genk
Hasselt
Aachen

Leeuwarden
Papingastins
NIEUW FRIESBURG
Heerenveen
Ambachtsstraat 5-3
6 hr 20 min
516 km
Zwolle
Westsingel 17-9
Kofferen 55
Dokkum
Drachten
Sneek
Den Helder
Alkmaar
Hoorn
Meppel
Amsterdam
Deventer
Apeldoorn
Leiden
Arnhem
Nijmegen
Dordrecht
's-Hertogenbosch
Breda
Tilburg
Eindhoven
Venlo


Roggestraat 19-27
Koorstraat 118
Kofferen 55
Truus Schröderstraat 2-10
Vinckenhofstraat 55-1
Beekstraat 54
Sint Wirosingel
4 hr 47 min
331 km
Minister Ruysstraat 8C
Akerstraat
Prins Bisschopsingel
Nijmegen
Kleve
Oss
Hertogenbosch
Uden
Veghel
Kevelaer
Best
Helmond
Deurne
Eindhoven
Sevenum
Valkenswaard
Nettetal
Lommel
Peer
Heinsberg
Erkelenz
Genk
Hasselt
Eschweiler
Aachen
Google


Broekweg 56-140
Polakweg 7
Schubertlaan
Gaffelstraat 92-64
Ecohoning
Zwaanwijk
4 hr 41 min
308 km
Koffer
Amstelveen
Lisse
Hilversum
Leiden
Amersfoort
Alphen aan den Rijn
Utrecht
Delft
Gouda
Nieuwegein
Tiel
Spijkenisse
Gorinchem
Haringvliet
Nationaal Park De Biesbosch
Oss
's-Hertogenbosch
Steenbergen
Breda
Tilburg
Roosendaal
Bergen op Zoom
Essen
Hilvarenbeek
Best
Eindhoven
Hoogstraten
Valkenswaard